Quark mass spectrum substrate decomposition attempt v0.1 — HONEST RESULT: quark masses NOT cleanly substrate-decomposable at lepton-precision level. Scheme-dependence + bulk-color mechanism open create barriers. CKM angles ARE clean ($\sin\theta$_C = N_c^2/(2^{N_c}·n_C) = 9/40 at 0.3%). Best quark-mass-ratio matches: m_b/m_c ≈ rank·n_C/N_c (1.2% off); m_t/m_b ≈ C_2·g-1 (0.6% off). Lepton-style closed forms blocked on bulk-color.

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Quark mass spectrum substrate decomposition attempt v0.1

0. Per Casey's directive sequence

After lepton masses (T190, T2003) substrate-decomposed and OTHER lepton observables (Schwinger a_e = 1/(2π·N_max), muon decay 192 = N_c·2^{C_2}) decomposed, the next item is quark mass spectrum.

1. The structural obstacle — scheme dependence

Quark masses are SCHEME-DEPENDENT in QCD: - MS̄-bar quark masses (running, scheme-defined). - Pole quark masses (gauge-invariant but with renormalon ambiguity). - Constituent quark masses (effective, model-dependent).

PDG values (MS̄-bar at appropriate scales): - m_u ≈ 2.16 MeV (at 2 GeV). - m_d ≈ 4.67 MeV. - m_s ≈ 93.4 MeV. - m_c ≈ 1.27 GeV. - m_b ≈ 4.18 GeV. - m_t ≈ 172.57 GeV (pole, more well-defined).

The scheme-dependence + the bulk-color mechanism still open create real barriers. Cal #27 from Friday + BST's quark sector scheme-invariance audit (#218) already flagged scheme-dependent quark mass leads as excluded from forward derivations.

2. CKM angles (scheme-invariant) — ARE substrate-clean

Cabibbo angle:

$\sin\theta$_C = N_c^2 / (2^{N_c} · n_C) = 9/40 = 0.225 vs observed 0.2243 (0.31% match)

Substrate-decomposition: 9 = N_c^2 (color squared); 40 = 2^{N_c} · n_C = 8 · 5 (Mersenne base raised to color, times complex dim). All substrate-primary.

This matches the pattern from lepton work: substrate primaries with rank/Mersenne base raised to primary exponents.

Other CKM angles also have substrate-natural fractions over N_max (PMNS-like; from F1 falsifier work): - |V_us| ≈ sin θ_C ≈ 0.225 = 9/40 = N_c^2/(2^{N_c}·n_C). Clean. - |V_cb| ≈ 0.041 (smaller). Substrate decomposition multi-week. - |V_ub| ≈ 0.004 (smallest). Substrate decomposition multi-week.

3. Quark mass-ratio attempts — best matches at 1-2% precision

Quark mass ratio	Observed	Best substrate candidate	Match
m_u/m_d	0.463	1/rank = 0.5	8% off
m_d/m_u	2.16	rank + 1/(rank+1)	not clean
m_s/m_d	20.0	n_C·g/rank·…	various 5-15%
m_c/m_s	13.6	(no clean match)	—
m_b/m_c	3.29	rank·n_C/N_c = 10/3 = 3.33	1.2% off
m_t/m_b	41.26	C_2·g - 1 = 41	0.6% off

The best matches (m_b/m_c, m_t/m_b) are at 1-2% precision — substantially worse than T190's 0.004% or T2003's 0.05%. The substrate-primary candidates are SUGGESTIVE but not PRECISE in the way lepton ratios are.

Possible reasons: 1. Scheme dependence: quark masses depend on renormalization scheme; substrate-natural derivation requires specifying scheme (typically MS̄-bar). The 1-2% precision might match the scheme-uncertainty. 2. Bulk-color mechanism: quarks are bulk K-types with color; their masses involve bulk-color structure not yet derived. Substrate-natural decomposition likely BLOCKED on bulk-color resolution. 3. Different mass mechanism: quarks may use different substrate-natural mechanisms than leptons (boundary vs bulk; different channel structures).

4. Cross-sector ratios (quark-lepton)

Cross-sector mass ratios involve substantially different mechanisms (boundary vs bulk).

- m_t/m_e ≈ 338,000. $\log_2$ ≈ 18.4. No clean substrate-primary integer match.
- m_b/m_e ≈ 8,180. $\log_2$ ≈ 13. No clean match.
- m_c/m_e ≈ 2,488. $\log_2$ ≈ 11.3. No clean match.

Cross-sector lepton-quark ratios don't fit a simple closed-form pattern. The lepton sector (T190, T2003 family) and quark sector use different substrate-natural mechanisms.

5. Substrate-natural quark observables identified so far

Observable	Substrate form	Match	Sector
sin θ_C (Cabibbo)	N_c^2/(2^{N_c}·n_C) = 9/40	0.3%	CKM (scheme-invariant)
m_b/m_c (suggestive)	rank·n_C/N_c = 10/3	1.2%	MS̄-bar (scheme-dependent)
m_t/m_b (suggestive)	C_2·g − 1 = 41	0.6%	MS̄-bar (scheme-dependent)
m_p/m_e (bulk composite)	C_2·π^{n_C} = 6π^5	0.002% (T187)	bulk composite

The CLEANEST quark-sector substrate decompositions are CKM angles + bulk composites (proton). Individual quark mass ratios show 1-2% suggestive matches but not lepton-precision clean closed forms.

6. What would unlock cleaner quark mass derivation

The bulk-color mechanism resolution (currently multi-week open, my v0.5+v0.6 work) is the LIKELY KEY: with explicit bulk-color SU(3)/Toeplitz structure, quark masses would have: - Color-octet gauge contributions (8-gluon adjoint). - Bulk K-type radial structure with color-dependent kernel integrals. - Substrate-natural exponents tied to color sector (vs. lepton's spinor sector).

Multi-week target after bulk-color closure verification.

7. Honest scope + tier

RIGOROUS (existing + arithmetic): - PDG quark masses (scheme-dependent). - Cabibbo angle decomposition $\sin\theta\_C = 9/40 = N\_c^2/(2\string^\{N\_c\}\cdot n\_C)$ at 0.3% (existing BST). - Various 1-2% suggestive matches for m_b/m_c, m_t/m_b. - $m\_p/m\_e = 6\pi^5$ (T187, bulk composite, clean).

HONEST FINDING (v0.1): quark mass RATIOS NOT cleanly substrate-decomposable at lepton-precision level. Best matches at 1-2% precision (m_b/m_c ≈ rank·n_C/N_c; m_t/m_b ≈ C_2·g − 1). CKM angles ($\sin\theta$_C) ARE clean. Bulk-color mechanism resolution likely needed for explicit quark mass derivation.

OPEN (multi-week): bulk-color mechanism resolution; scheme-invariance handling; explicit quark mass derivation via bulk K-type structure with color.

Cal #27 / honesty: NOT claiming quark mass ratios are substrate-decomposed. Honest finding: lepton-precision closed forms NOT available for quarks at v0.1; scheme-dependence + bulk-color blocking. Suggestive 1-2% matches noted as candidates for further investigation but NOT promoted to clean closed forms. CKM angles ARE clean (per existing BST). The structural picture: lepton sector and quark sector have different substrate-natural mechanisms; quark sector mechanisms blocked on bulk-color.

Routed: → Casey: quark mass derivation is the next major depth target but is blocked on bulk-color mechanism resolution. CKM angles cleanly substrate-derived. Suggestive m_b/m_c ≈ rank·n_C/N_c and m_t/m_b ≈ C_2·g − 1 noted but unpromoted. → Elie: when bulk-color mechanism resolves (multi-week), quark mass derivations via bulk K-type radial tower + color structure become accessible. → Grace: catalog candidate m_b/m_c ≈ 10/3 and m_t/m_b ≈ 41 as suggestive substrate candidates with 1-2% precision (lower-tier than lepton closed forms). → me: next per Casey's "keep going" — hadron spectrum or gauge bosons.

— Lyra, quark mass spectrum v0.1. HONEST FINDING: quark mass ratios NOT cleanly substrate-decomposable at lepton-precision level. Scheme-dependence + bulk-color open create barriers. CKM angles ARE clean ($\sin\theta\_C = 9/40 = N\_c^2/(2\string^\{N\_c\}\cdot n\_C)$, 0.3%). Suggestive 1-2% matches for m_b/m_c ≈ rank·n_C/N_c and m_t/m_b ≈ C_2·g−1 noted as candidates. Lepton-style closed forms blocked on bulk-color resolution; multi-week target.